

Lecture 15: Plane wave propagation

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Outline and Motivation

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- For the special case of **isotropic** materials, we will derive the familiar P- and S-waves.
- We will then use perturbation theory to see how these waves are affected by slight anisotropy.
- Finally, we will qualitatively discuss how plane waves interact with boundaries (reflection and conversion).

The Setup: Plane Waves in a Homogeneous Medium

We start with the linearised equations of motion for a **homogeneous** body, where ρ and A_{ijkl} are constant:

$$\rho \frac{\partial^2 u_i}{\partial t^2} - A_{ijkl} \frac{\partial^2 u_k}{\partial x_j \partial x_l} = 0$$

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Slowness Vector ($\mathbf{p}$)

- Direction of $\mathbf{p}$ is the propagation direction.
- Magnitude sets the phase speed, $c = 1/||\mathbf{p}||$.

Polarisation Vector ($\mathbf{a}$)

- A unit vector describing the direction of particle motion.

Deriving the Christoffel Equation

We substitute the plane wave form $u_i = f(t - \mathbf{p} \cdot \mathbf{x})a_i$ into the equation of motion.

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Rearranging this gives the fundamental condition for a plane wave to exist.

The Christoffel Equation

The Christoffel Matrix

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The condition for plane wave solutions can now be written compactly as:

$$\Gamma_{ik} a_k = a_i \quad \text{or} \quad [\mathbf{\Gamma}(\mathbf{p}) - \mathbf{I}] \mathbf{a} = 0$$

This is a homogeneous system of linear equations for the polarisation vector $\mathbf{a}$.

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This is a homogeneous system of linear equations for the polarisation vector $\mathbf{a}$.

- For a non-trivial solution ($\mathbf{a} \neq 0$) to exist, the determinant of the matrix must be zero:

$$\det[\mathbf{\Gamma}(\mathbf{p}) - \mathbf{I}] = 0$$

Reduction to a Symmetric Eigenvalue Problem

We can rewrite the Christoffel equation in a more familiar form.

- Let the slowness vector be $\mathbf{p} = \frac{1}{c}\hat{\mathbf{p}}$, where c is the phase speed and $\hat{\mathbf{p}}$ is the direction.

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- The Christoffel matrix has the property $\mathbf{\Gamma}(\lambda\mathbf{p}) = \lambda^2\mathbf{\Gamma}(\mathbf{p})$. Using this, the Christoffel equation becomes:

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Symmetry

Using the hyperelastic symmetry ($A_{ijkl} = A_{klij}$), one can show that the Christoffel matrix $\mathbf{\Gamma}(\hat{\mathbf{p}})$ is always symmetric.

Qualitative Solution

Because $\mathbf{\Gamma}(\hat{\mathbf{p}})$ is a 3×3 symmetric matrix, we know the properties of its solutions from linear algebra.

- It has **three real eigenvalues**: c_1^2, c_2^2, c_3^2 .

Qualitative Solution

Because $\mathbf{\Gamma}(\hat{\mathbf{p}})$ is a 3×3 symmetric matrix, we know the properties of its solutions from linear algebra.

- It has **three real eigenvalues**: c_1^2, c_2^2, c_3^2 .
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Physical Interpretation

In any given propagation direction $\hat{\mathbf{p}}$, there are (up to) **three distinct types of plane waves**.

- Each wave type has its own phase speed (c_1, c_2, c_3).
- The particle motions (polarisations $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$) of these three waves are mutually orthogonal.

Isotropic Case: P- and S-Waves

For an isotropic material, we can solve the eigenvalue problem explicitly. The Christoffel matrix becomes:

$$\Gamma_{ik}(\hat{\mathbf{p}}) = \frac{\lambda + \mu}{\rho} \hat{p}_i \hat{p}_k + \frac{\mu}{\rho} \delta_{ik}$$

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We can immediately find the eigenvectors and eigenvalues.

P-waves (Longitudinal)

If we choose the polarisation vector parallel to the propagation direction, $\mathbf{a} = \hat{\mathbf{p}}$:

$$\Gamma(\hat{\mathbf{p}})\hat{\mathbf{p}} = \left(\frac{\lambda + 2\mu}{\rho} \right) \hat{\mathbf{p}}$$

- This is a longitudinal wave, called a **P-wave**.
- The phase speed is $c_P = \alpha = \sqrt{(\lambda + 2\mu)/\rho}$.

Isotropic Case: S-Waves and Summary

S-waves (Transverse)

If we choose $\mathbf{a}$ to be any vector orthogonal to $\hat{\mathbf{p}}$ (i.e., $\mathbf{a} \cdot \hat{\mathbf{p}} = 0$):

$$\mathbf{\Gamma}(\hat{\mathbf{p}})\mathbf{a} = \left(\frac{\mu}{\rho}\right)\mathbf{a}$$

- This is a transverse wave, called an **S-wave**.
- The phase speed is $c_S = \beta = \sqrt{\mu/\rho}$.
- Since there are two orthogonal directions transverse to $\hat{\mathbf{p}}$, this eigenvalue is **two-fold degenerate**.

Isotropic Case: S-Waves and Summary

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Summary for Isotropic Media

- One type of P-wave with speed α .
- Two types of S-waves with the same speed β .
- Speeds are independent of propagation direction.

Slightly Anisotropic Materials

In some parts of the Earth, the material is slightly anisotropic. We can analyze this using perturbation theory.

- We write the elastic tensor as an isotropic part plus a small anisotropic perturbation:

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- We then find the first-order corrections to the isotropic eigenvalues (c^2) and eigenvectors ($\mathbf{a}$).

Quasi P-wave

The non-degenerate P-wave is perturbed slightly.

- The wave is now a **quasi P-wave** (qP).
- The polarisation is almost, but not exactly, parallel to $\hat{\mathbf{p}}$.
- The phase speed now depends on the propagation direction.

Perturbation of P- and S-waves

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Quasi S-waves and Shear-Wave Splitting

The two-fold S-wave degeneracy is the most interesting case.

- Perturbation theory shows that the degeneracy is almost always lifted. The two S-waves now have slightly different speeds.
- This phenomenon is called **shear-wave splitting**.
- The resulting waves are **quasi S-waves** (qS1, qS2) with nearly transverse, orthogonal polarisations.

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Geophysical Significance of Shear-Wave Splitting

Probing Mantle Flow

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- This is caused by the large-scale alignment of mineral grains due to convective flow.
- By observing shear-wave splitting, seismologists can make inferences about the direction and nature of flow within the Earth's mantle.

Waves at an Interface: Snell's Law

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Snell's Law

This phase-matching condition requires that the component of the slowness vector parallel to the interface must be conserved for all incident, reflected, and transmitted waves.

$$p_{\text{horizontal}} = \text{constant}$$

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Complexity of Seismograms

Wave conversion is a fundamental property of elasticity and is a major reason why observed seismograms are so complex, containing many more arrivals than just the direct P- and S-waves.

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- How to solve the Christoffel equation for an isotropic solid to find **P-waves** (longitudinal) and **S-waves** (transverse, degenerate).
- The effect of slight anisotropy: P-waves become **quasi-P**, and S-waves exhibit **shear-wave splitting**.
- The basics of wave interaction at a boundary: **Snell's Law** (conservation of horizontal slowness) and **wave conversion**.